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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

EVENT REGISTRATION SYSTEM

PROJECT REPORT

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS (HONOURS)

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DECLARATION

I hereby declare that the project report entitled “EVENT REGISTRATION SYSTEM”, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (Honours) at RV University, is a record of work carried out by me under the guidance of Pranjul Shukla.

I further declare that this work has not been submitted, either in part or in full, for the award of any other degree or diploma at this or any other institution.

Place: Bengaluru
Date: [02/11/2026]

Signature of the Candidate
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CERTIFICATE

This is to certify that the project report entitled “EVENT REGISTRATION SYSTEM” is a bona fide record of the work carried out by SACHITH N (USN: 1RUA25BCA0092) under my guidance, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (Honours) at RV University.

The work presented in this report has been carried out during the academic year 2026–2027.

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Signature of Guide	Signature of Center Head /Dean
Designation, SOCSE, RV University	SOCSE, RV University

ABSTRACT

The Event Registration System is a desktop application developed using Java Swing and MySQL. It provides a simple interface to manage participant registrations for events. The application stores event details and participant information in a relational database and connects to MySQL using Java Database Connectivity (JDBC).

The database contains two main tables: events and registrations. Users can select an event, enter a participant's name and phone number, view existing registrations, update selected records and delete records. The application uses SQL queries, prepared statements and a join to display registration details along with event names and dates.

This project demonstrates the integration of Java GUI programming with database management concepts such as primary keys, foreign keys, joins and CRUD operations.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Events such as workshops, seminars, competitions and college festivals require organisers to maintain participant details. Manual registration through paper forms or separate spreadsheets can make it difficult to search, update and organise records.

The Event Registration System provides a simple computerised solution. It allows the user to enter participant information, select an event and manage registrations through a desktop interface. The information is stored in a MySQL database so that it remains available after the application is closed.

1.2 AIM

To develop a Java-based desktop application connected to a MySQL database for storing and managing event registrations.

1.3 OBJECTIVES

- Create a database to store event and participant details.
- Develop a Java Swing interface for registration management.
- Connect the application to MySQL using JDBC.
- Implement adding, viewing, updating and deleting registrations.
- Display participant details together with the selected event.

1.4 PROBLEM STATEMENT

The project addresses the need for a centralised and structured way to record participants and associate each participant with an event. The application reduces reliance on manual record maintenance for a small event-registration workflow.

1.5 SCOPE OF THE PROJECT

The current project is a desktop application intended for basic event-registration management. It supports the following functions:

- Retrieve event names and dates from the database.
- Enter a participant name and phone number.
- Choose an event from a drop-down list.
- View existing registrations in a table.
- Update or delete a selected registration.

1.6 LIMITATIONS

- The application is designed for local desktop use and does not provide online registration.
- It does not include user login, payment, email notifications or QR-code tickets.
- It does not currently provide event capacity management or attendance tracking.

1.7 REPORT ORGANISATION

Chapter 2 describes the technologies and database concepts. Chapter 3 explains the methodology and database design. Chapter 4 presents the implementation. Chapter 5 summarises the sample data and testing approach. Chapter 6 discusses findings and limitations, followed by the conclusion, references and appendix.

CHAPTER 2

BACKGROUND

2.1 OVERVIEW

An event registration process collects participant information and records the event for which the person has registered. In this project, the required information is limited to the participant's name, phone number and selected event.

2.2 TECHNOLOGIES USED

Technology	Purpose
Java	Application programming language.
Java Swing	Creates the desktop graphical user interface.
JDBC	Connects Java to the MySQL database.
MySQL	Stores event and registration records.
NetBeans / Java IDE	Used to develop and run the Java application.

2.3 DATABASE CONCEPTS

A relational database stores information in tables. A primary key uniquely identifies each row. A foreign key connects records in related tables. In this project, each registration refers to an event through event_id.

2.4 CRUD OPERATIONS

CRUD refers to Create, Read, Update and Delete. These are the basic operations used to manage the registration records. The application implements these operations through SQL statements executed using JDBC.

CHAPTER 3

METHODOLOGY

3.1 DEVELOPMENT METHODOLOGY

The project was developed in stages. First, the information required for registration was identified. Next, the MySQL database and tables were created. The Java Swing interface was then designed, followed by JDBC integration and implementation of the database operations.

1. Identify the event and participant information to be stored.
2. Create the database and tables in MySQL.
3. Insert sample event and registration records.
4. Build the Java Swing user interface.
5. Connect the application to MySQL using JDBC.
6. Implement and verify the registration operations.

3.2 SYSTEM REQUIREMENTS

Hardware / Software	**Requirement**
Computer	Desktop or laptop capable of running Java and MySQL.
Operating system	Windows or another Java-supported operating system.
Programming platform	Java Development Kit (JDK).
Database	MySQL Server and MySQL Workbench.
Driver	MySQL Connector/J on the Java classpath.

3.3 DATABASE DESIGN

The database is named event_registrations_db and contains two tables: events and registrations. The events table stores event details. The registrations table stores participant details and the event identifier.

Table / Column	Data type	Description
events.event_id	INT	Primary key; auto-increment.
events.event_name	VARCHAR(100)	Name of the event.
events.event_date	DATE	Date of the event.
registrations.registration_id	INT	Primary key; auto-increment.
registrations.registor_name	VARCHAR(100)	Participant name.
registrations.phone	VARCHAR(100)	Phone number.
registrations.event_id	INT	Foreign key referencing events.

3.4 RELATIONSHIP

One event can have multiple registrations. Each registration is associated with one event. The relationship is implemented through the event_id foreign key in the registrations table.

EVENTS (1) ———————— (M) REGISTRATIONS

CHAPTER 4

IMPLEMENTATION

4.1 USER INTERFACE

The application is implemented in a Java class named EventRegistrationApp. It extends JFrame and creates the main application window. The interface contains text fields for participant name and phone number, a combo box for event selection, buttons for adding and refreshing records, and a table for displaying registrations.

4.2 DATABASE CONNECTION

The connect() method uses DriverManager.getConnection() to establish a JDBC connection to MySQL. The application uses the database URL, username and password to connect to event_registrations_db. The MySQL Connector/J driver must be available in the project classpath.

4.3 LOADING EVENTS

The loadEvents() method retrieves event IDs, names and dates from the events table. The results are used to populate the event-selection combo box.

4.4 DISPLAYING REGISTRATIONS

The loadRegistrations() method retrieves registration information by joining the registrations table with the events table. The resulting rows are displayed in a JTable.

4.5 ADDING A REGISTRATION

The user enters a name and phone number and selects an event. The application checks that the required fields are not empty and that the text length does not exceed the database column limit. It then executes an INSERT statement using PreparedStatement.

4.6 UPDATING A REGISTRATION

To update a record, the user selects a row in the table. The selected participant details are placed into the form. After editing the values, the Update operation modifies the selected registration using its registration_id.

4.7 DELETING A REGISTRATION

The user selects a registration and chooses the delete operation. A confirmation message is displayed before the record is deleted from the database.

4.8 REFRESHING DATA

The refresh function reloads the event list and registration table so that the interface displays the latest database records.

4.9 ERROR HANDLING

The application checks for missing input and displays messages for database errors. Try-with-resources is used for database objects so that connections and statements are closed after use.

CHAPTER 5

RESULTS AND TESTING

5.1 SAMPLE DATA

The SQL script inserts three sample events and three sample registrations. These records can be used to check whether the application retrieves and displays data from MySQL.

Event	**Date**
CodeCon	21-10-2026
Argonyx	23-11-2026
AR-VR	25-11-2026
Participant	**Event**
Rahul Sharma	CodeCon
Priya Nair	Argonyx
Arjun Rao	AR-VR

5.2 EXPECTED FUNCTIONAL RESULTS

- Events should appear in the event-selection list.
- Existing registrations should appear in the table with event details.
- A valid new registration should be stored in MySQL.
- An updated registration should display the changed details.
- A deleted registration should no longer appear in the list.

5.3 TESTING

The following tests should be performed on the local system. Record the actual outcome after running the application.

Test	**Action**	**Expected result**	**Actual result**
1	Launch application	Window and records load	[Fill]
2	Add valid registration	Record is saved	[Fill]
3	Submit empty fields	Validation message appears	[Fill]
4	Update selected record	Record is modified	[Fill]
5	Delete selected record	Record is removed	[Fill]
6	Refresh list	Latest data appears	[Fill]

The source repository contains the application code and SQL script, but does not include screenshots or a completed test log. Add screenshots from the actual running application before submission.

CHAPTER 6

INFERENCES

6.1 KEY FINDINGS

- The project demonstrates Java Swing integration with a MySQL database.
- The two-table design separates event details from participant registrations.
- Primary and foreign keys establish the relationship between the tables.
- JDBC and PreparedStatement are used to execute database operations.
- The application supports the basic CRUD operations for registrations.

6.2 LIMITATIONS

The current version is a basic desktop application. It does not include online registration, authentication, payment processing, notifications, event capacity limits or attendance tracking. The application also requires a working local MySQL server and correct JDBC configuration.

6.3 SECURITY NOTE

Database credentials should not be stored directly in source code or published in a public repository. Use environment variables or a protected configuration file and a database account with limited privileges.

DATABASE QUERY

```
CREATE DATABASE IF NOT EXISTS event_registrations_db
CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE event_registrations_db;

CREATE TABLE IF NOT EXISTS events (
  event_id INT PRIMARY KEY AUTO_INCREMENT,
  event_name VARCHAR(100) NOT NULL,
  event_date DATE NOT NULL
);

CREATE TABLE IF NOT EXISTS registrations (
  registration_id INT PRIMARY KEY AUTO_INCREMENT,
  registor_name VARCHAR(100) NOT NULL,
  phone VARCHAR(100) NOT NULL,
  event_id INT NOT NULL,
  CONSTRAINT fk_registrations_event FOREIGN KEY (event_id)
    REFERENCES events(event_id)
);

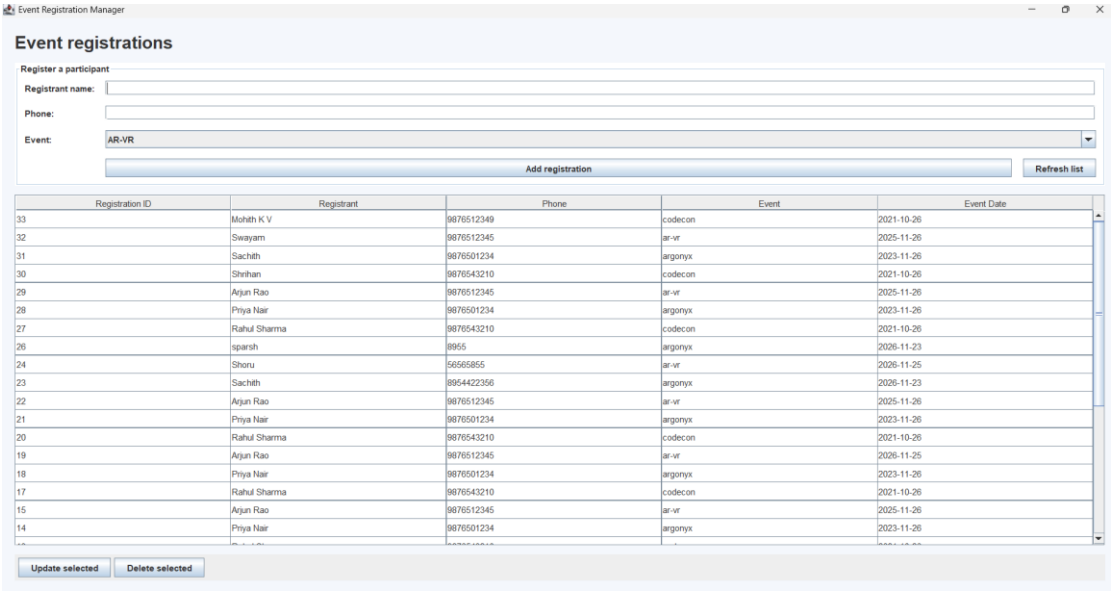
-- MySQL DATE values use YYYY-MM-DD format.
INSERT INTO events (event_name, event_date) VALUES
 ('codecon', '2026-10-21'),
 ('argonyx', '2026-11-23'),
 ('ar-vr', '2026-11-25');
```

```
INSERT INTO registrations (registor_name, phone, event_id) VALUES
 ('Rahul Sharma', '9876543210', 1),
 ('Priya Nair', '9876501234', 2),
 ('Arjun Rao', '9876512345', 3);

-- View events
SELECT * FROM events;

-- View registrations
SELECT * FROM registrations;
```

APPLICATION



CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The Event Registration System provides a simple way to store and manage event registrations using Java Swing and MySQL. It connects a graphical interface to a relational database through JDBC and implements adding, viewing, updating and deleting participant records. The project demonstrates practical use of database tables, keys, joins and prepared statements.

7.2 FUTURE WORK

- Develop a web-based version for online registration.
- Add administrator login and access control.
- Include event capacity and registration deadline management.
- Add email or SMS confirmations.
- Generate event-wise participant reports.
- Introduce QR-code tickets and attendance tracking.

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GitHub Link - https://github.com/sachithn-27/DBMS_CIE3.git